AIN SHAMS UNIVERSITY
FACULTY OF ENGINEERING

Computer Engineering and Software Systems Program
(Senior Level), Computer and Systems Engineering

Fall, 2025/2026 **Course Code: CSE381** **Time allowed: 1 Hr.**

Introduction to Machine Learning (Mid Term Mock Exam)

The Exam Consists of **Two** Questions in **Three** Pages. **Maximum Marks: 20 Marks**

Important Rules:
- Having a mobile, Smart Watch or earphones inside the examination hall is forbidden and is considered as a cheating behavior.
- It is forbidden to have any references, notes, books, or any other materials even if it is not related to the exam content with you in the examination hall.

تعليمات هامة
- حيازة (المحمول- الساعات الذكية - سماعة الأذن) داخل لجنة الامتحان يعتبر حالة غش تستوجب العقاب .
- لايسمح بدخول أي كتب أو ملازم أو أوراق داخل اللجنة والمخالفة تعتبر حالة غش.

Student Name: Student ID:

Question (1): **[9 marks]**

I. Match:

(i) Given a high-dimension small dataset, efficient visualization is required.	A. Supervised learning.
(ii) Given a dataset of patients diagnosed as either normal or severe, learn to classify new classify new patients as normal or not.	B. Unsupervised learning.
(iii) Given a dataset of faces of known persons, the identity corresponding to a new face is required.	C. None of these choices.
(iv) Given a large electronic market of identical products, it is required to estimate how many products will be sold in the next month.	

[0.5 mark each]

II. Match:

(i) Given a high-dimension small dataset, efficient visualization is required.	A. Regression.
(ii) Given a dataset of patients diagnosed as either normal or severe, learn to classify new classify new patients as normal or not.	B. Classification.
(iii) Given a dataset of faces of known persons, the identity corresponding to a new face is required.	C. Clustering.
(iv) Given a large electronic market of identical products, it is required to estimate how many products will be sold in the next month.	D. Dimensionality reduction.
	E. None of these choices.

[0.5 mark each]

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III. Tick (√) or (×) and correct the false: [0.5 mark each]

(a) Clustering refers to techniques that reduce the number of input variables in a dataset.

(b) Dimensionality reduction techniques are often used for data visualization.

(c) Reinforcement learning includes robot navigation.

(d) For rapid convergence and good generalization, it is better to scale features.

(e) Regularization can be implemented via adding a term to the cost function that can guarantee small values for the coefficients.

(f) Analytical solution of regression is not suitable for large number of features.

IV. Choose the correct answer: [0.5 mark each]

(a) The hypothesis function $H(\Theta_0, \Theta_1) = \Theta_0 + \Theta_1 x + \Theta_2 x^2$ is considered as:

- Linear regression, Polynomial regression, All these choices, or None of these choices.

(b) What is the main reason of the overfitting?

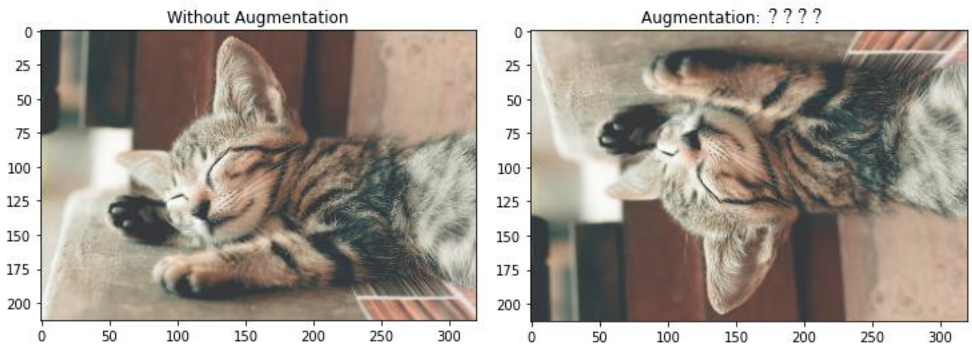
- Very small dataset, simple model, All these choices, or None of these choices.

(c) When the model exactly fits the training data but cannot predict unseen data, this case is called:

- Underfitting, Overfitting, All these choices, or None of these choices.

(d) What is the type of image augmentation applied in the following case?

- Horizontal flip, vertical flip, rotation, zoom in, height shift, lighting transformation, or None of these choices.



<u>Question (2)</u>: [11 marks]

(a) It is required to build an OCR. The input image contains four Japanese characters:

月 目 木 山

The extracted feature is the number of holes (H). Compute the number of holes (H) of each of these characters. Draw the feature map in this case. Are theses classes separable? Is it possible to classifiy the character using a single feature (H)? Justify your answer. [2 marks]

(b) Apply the gradient descent algorithm for the following regression problem with the following data (x, y): (0.3, 5), (1.4, 18), (2.1, 37). The hypothesis function $H(\Theta_0, \Theta_1) = \Theta_0 + \Theta_1 x$, with initial values for Θ_0 and Θ_1 = 0.1 and 0.33, respectively. The learning rate is 0.85. Show the results after the first iteration only.

<u>Useful Hints:</u>

Cost Function:

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$$

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repeat until convergence {

$$\theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right)$$

$$\theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right) \cdot x^{(i)}$$

}

[5 marks]

(c) For features ***x1*** and ***x2***, where the range of ***x1*** is an integer between ***65*** and ***70***, and the range of ***x2*** is an integer between ***3*** and ***6***. Apply each of the following scaling techniques:

(i) MIN-MAX normalization.

$$X'_i = \frac{x_i - \min(x)}{\max(x) - \min(x)}$$

(ii) Z Score normalization.

$$x_i = \frac{x_i - \mu_i}{\sigma_i}$$

(μ_i *is the mean of* x_i *training set*)

σ_i *is the standard deviation of* x_i *training set*)

standard deviation

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{n}}$$

Find the new features after scaling in each case and compare between them. [4 marks]

END of Exam, Good Luck

Examination Committee *Dr. Alaa Hamdy* ***Exam. Date : 8th of November, 2025***